

Answer all questions.

1. (a) In the production of steel alloy, atoms of carbon are added to iron. The resulting alloy is less *ductile* than pure iron. State the meaning of the term *ductile*, and describe, on an atomic scale, why the addition of carbon atoms can make steel less ductile than iron. [3]

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- (b) A wire of length 2.4m and diameter 0.60mm is made of steel of Young modulus $200 \times 10^9 \text{ Nm}^{-2}$. The wire is loaded so that its length is increased by 1.8mm. Assuming that the change is elastic, calculate:

(i) the strain; [1]

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(ii) the applied stress; [2]

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(iii) the force applied to the wire; [2]

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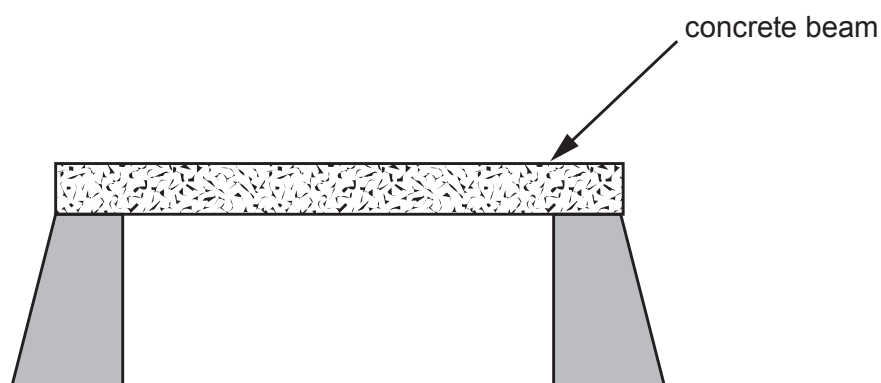
- (iv) the elastic energy stored in the wire. [2]

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- (c) The diagram shows how a gap can be bridged using a concrete beam.



- (i) **On the diagram** label a point in tension with the letter **T**, and a point in compression with the letter **C**. [1]
- (ii) Inserting a pre-stressed steel bar into the concrete beam would increase the breaking stress of the concrete. **On the diagram**, draw a pre-stressed steel bar in an appropriate position. [1]
- (iii) Explain how the steel bar strengthens the beam. [2]

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